

PROJECT REPORT

Optimizing User, Group, and Role Management with Access Control and Workflows

Submitted By

Srinidhi S

CSE

Sri Venkateswara College of Engineering

Academic Year: 2023–2027

Semester: IV-I

CERTIFICATE

This is to certify that Srinidhi S completed the ServiceNow project titled 'Optimizing User, Group, and Role Management with Access Control and Workflows' as partial fulfillment of the ServiceNow Administrator .

ACKNOWLEDGEMENT

I thank the Department of CSE, faculty members, friends, and my family for their support.

ABSTRACT

This ServiceNow application automates user, group, and role management using ACLs and Flow Designer workflows. It supports secure project management, approvals, and task automation.

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CHAPTER 1: INTRODUCTION

ServiceNow is a cloud-based enterprise platform used to automate business processes and manage digital workflows. This project focuses on optimizing user, group, and role management through secure access control and automated workflows. The application provides centralized management of projects and tasks while improving security using Access Control Lists (ACLs) and role-based permissions.

CHAPTER 2: LITERATURE SURVEY

Enterprise workflow systems improve productivity by automating repetitive processes. ServiceNow provides Flow Designer, custom tables, approvals, notifications, and security features that reduce manual effort. Previous approaches relied on manual task tracking, whereas workflow automation improves transparency, consistency, and accountability.

CHAPTER 3: PROBLEM STATEMENT

Organizations often face challenges in managing user permissions, project tasks, and approvals manually. Lack of centralized access control may result in security issues and inefficient collaboration. A secure automated solution is required to streamline these operations.

OBJECTIVES

- Develop a ServiceNow application for project management.
- Implement role-based access using ACLs.
- Automate task workflows using Flow Designer.
- Generate approval requests automatically.
- Improve security and collaboration.

EXISTING SYSTEM

Traditional project management involves manual assignment of tasks and approvals. Role management is often decentralized, leading to inconsistent permissions, delayed approvals, and increased administrative effort.

PROPOSED SYSTEM

The proposed system automates project and task management using ServiceNow. Project Managers create projects and tasks, Team Members perform assigned work, and Flow Designer automates approvals and status updates. ACLs ensure that users access only authorized records and fields.

FUNCTIONAL REQUIREMENTS

1. User and role management.
2. Project creation.
3. Task creation and assignment.
4. Workflow automation.
5. Approval management.
6. ACL configuration.
7. Role-based access.

NON-FUNCTIONAL REQUIREMENTS

- Security
- Reliability
- Scalability
- Performance
- Availability
- Usability
- Maintainability

CHAPTER 4: SYSTEM ARCHITECTURE

The application is developed on the ServiceNow platform using custom tables, Flow Designer, user roles, groups, and Access Control Lists (ACLs). Users interact with the application through the ServiceNow interface. Business logic is executed through Flow Designer while data is stored in custom tables.

Architecture Components

- ServiceNow Personal Developer Instance (PDI)
- Application Navigator
- Project Table
- Task Table 2
- Flow Designer

- ACLs
- Users and Groups
- Roles
- Approval Engine

Database Design

Project Table stores project details such as project name, manager, start date, end date, and status.

Task Table 2 stores task details including task name, assigned user, description, comments, priority, and task status.

User Roles

Administrator:

- Configure application
- Manage users and roles
- Configure ACLs

Project Manager:

- Create projects
- Create tasks
- Assign work
- Approve requests

Team Member:

- View assigned tasks
- Update permitted fields
- Participate in workflow

Access Control Lists (ACLs)

ACLs are configured on Task Table 2 and its fields to ensure that only authorized users can read, write, create, or modify records. Team Member role is granted controlled access while administrative functions remain restricted.

Workflow Design

The Flow Designer workflow starts when a new task record is created. The flow updates the task status, generates an approval request for the Project Manager, waits for approval, and completes the workflow after the approval is granted.

Implementation

Step 1: Create Project Table.

Step 2: Create Task Table 2.

Step 3: Create Project Manager and Team Member roles.

Step 4: Configure users and groups.

Step 5: Build Flow Designer workflow.

Step 6: Configure ACLs.

Step 7: Publish and activate the flow.

Step 8: Test the workflow.

CHAPTER 5: TESTING

The application was validated using Functional Testing, User Acceptance Testing (UAT), Workflow Testing, ACL Testing, Approval Testing, and Performance Testing. All planned test cases were executed successfully.

Functional Testing

Major functions tested include project creation, task creation, task assignment, workflow execution, approval generation, status updates, and role-based access control. Expected and actual results matched successfully.

User Acceptance Testing

Project Managers, Team Members, and Administrators verified that the application satisfied the specified business requirements. All acceptance criteria were achieved.

Performance Testing

Workflow execution completed successfully within acceptable response time. Role validation, approvals, and status updates were performed without errors. Overall system performance was satisfactory.

Bug Tracking

The major issues identified during development included workflow trigger configuration, task status updates, approval generation, and ACL permissions. All identified bugs were resolved before deployment.

Results and Discussion

The developed ServiceNow application successfully automated project management processes. The integration of Flow Designer with ACLs improved operational efficiency, reduced manual effort, and ensured secure access to project data.

Advantages

- Automated workflow execution
- Secure role-based access
- Centralized project management
- Reduced manual intervention

- Improved collaboration
- Better task tracking

Limitations

- Requires ServiceNow platform access
- Depends on correct ACL configuration
- Limited to configured workflows
- Requires administrator privileges for configuration

Future Enhancements

- Email notifications
- Mobile application support
- AI-assisted task assignment
- SLA monitoring
- Dashboard analytics
- Integration with Microsoft Teams and external services

Conclusion

The project demonstrates how ServiceNow can be used to build secure enterprise workflow applications. By combining custom tables, Flow Designer, approval workflows, and ACLs, the system provides an efficient solution for managing users, groups, roles, and project activities.

References

1. ServiceNow Product Documentation
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3. ServiceNow Flow Designer Documentation
4. ServiceNow Access Control Documentation
5. Agile Project Management Concepts

GitHub Repository

Repository Name: Optimizing User, Group, and Role Management with Access Control and Workflows